Reference Material

DS-RX1 Printer Driver “Print Direction” setting

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1. Summary

This material describes the “Print Direction” setting for the DS-RX1 printer driver.

2. About new media size and “Print Direction”

For the DS-RX1 printer driver, a new media type has been added to the DS40-compatible media types. Compared with the compatible media types, the new media type has a different “Print Orientation” in the driver settings. The following chart shows the relationship between the application preview display and the actual print of the original image, when the “Print Orientation” driver setting is changed for the compatible and new media.

	Original Image	Driver Setting “Print Orientation”	Explorer (Windows XP)		Adobe PhotoShop 7.0	
			Preview	Output Image (direction↓)	Preview	Output Image (direction↓)
DS40 Compatible media (5x3.5) (6x4) (6x4)x2		Portrait				
		Landscape				
		Portrait				
		Landscape				
DS-RX1 New media PR (3.5x5) PR (4x6) PR (4x6)x2		Portrait				
		Landscape				
		Portrait				
		Landscape				

3. How to interface with the driver

The image height x width sizes for the compatible and new media types are opposite.
When creating an application, the differences between the compatible and new media are as follows.

3.1 Compatible media

The interface can be made the same as with the DS40.

3.2 New media

The following two methods can be used.

a) Media orientation parameter setting

Setting the media orientation using the member dmOrientation of the structure DEVMODEW.
Refer to the DEVMODEW structure members in the SDK document “Print Settings using the Application”.

Original Image	Media	DEVMODE structure member Setting the member dmOrientation	Output Image
	Compatible (5x3.5) (6x4) (6x4)x2	**DM_LANDSCAPE**	
	New PR (3.5x5) PR (4x6) PR (4x6)x2	**DM_PORTRAIT**	
	compatible (5x3.5) (6x4) (6x4)x2	**DM_PORTRAIT**	
	New PR (3.5x5) PR (4x6) PR (4x6)x2	**DM_LANDSCAPE**	

* Separate enlargement/reduction is necessary for fitting the image to the media size.

b) Rotate the image using the application

Rotate the image using the application by determining the image and media height/width sizes, while leaving the driver “Print Orientation” setting in the “Portrait” default.

Original Image	Media	Rotate the image using the application	Output Image
	Compatible (5x3.5) (6x4) (6x4)x2	**Necessary (90° or 270°)**	
	New PR (3.5x5) PR (4x6) PR (4x6)x2	**Unnecessary**	
	Compatible (5x3.5) (6x4) (6x4)x2	**Unnecessary**	
	New PR (3.5x5) PR (4x6) PR (4x6)x2	**Necessary (90° or 270°)**	

* Separate enlargement/reduction is necessary for fitting the image to the media size.

4. Image Size

Media Size	Orientation	Image size(Width x height)		Image pixel (width x height)		
		(inch)	(mm)	300x300 dpi	300x600 dpi	600x600 dpi
(5x3.5)	Portrait	5”x 3.5”	131.1 x 92.1	1548 x 1088	1548 x 2176	3096 x 2176
	Landscape	3.5”x 5”	92.1 x 131.1	1088 x 1548	2176 x 1548	2176 x 3096
(6x4) (6x4)x2	Portrait	6”x 4”	156.1 x 105.0	1844 x 1240	1844 x 2480	3688 x 2480
	Landscape	4”x 6”	105.1 x 156.1	1240 x 1844	2480 x 1844	2480 x 3688
PR(3.5x5)	Portrait	3.5”x 5”	92.1 x 131.1	1088 x 1548	2176 x 1548	2176 x 3096
	Landscape	5”x 3.5”	131.1 x 92.1	1548 x 1088	1548 x 2176	3096 x 2176
PR(4x6) PR(4x6)x2	Portrait	4”x 6”	105.1 x 156.1	1240 x 1844	2480 x 1844	2480 x 3688
	Landscape	6”x 4”	156.1 x 105.0	1844 x 1240	1844 x 2480	3688 x 2480